

# Ruiding Feng

## RESEARCH INTERESTS

My research interests span computer vision, computer graphics, and robotics, with an emphasis on physical AI. I am especially interested in structured object and scene generation, articulated and kinematic 3D asset modeling, physics-aware simulation, real-to-sim reconstruction, and policy learning for embodied systems.

## EDUCATION BACKGROUND

### **Rutgers University**

01/2026 - 05/2027

- Expected degree: MS in Computer Science | New Brunswick, NJ
- Main Courses: Machine Learning, Computer Vision, Data Structures, Mathematics for Computer Science, Computer Graphics, Natural Language Processing

### **SCAD**

09/2023 - 03/2025

- Software: Houdini, Nuke, UE5, Maya, Zbrush | Savannah, GA
- Core Skills: VEX scripting, FX simulation, Compositing, Lighting, Modeling etc.

### **Boston University (BA)**

08/2021 - 08/2023

- Degree: BA in Mathematics & Computer Science | Boston, MA
- Main Courses: Data Structures, Algorithms, Computer Systems, Software Engineering, Applied Statistics, Calculus & Multivariate Calculus, Probability, Stochastic Processes

## PUBLICATIONS AND MANUSCRIPTS

### **TreeStruct3D: Enabling Structural Editability in Agentic Procedural 3D Modeling**

**Ruiding Feng**, Sen Zhang, Danrui Li, Bingjiang Xia, Zhijin Zhu, Mubbasir Kapadia

*Under review by 3DV 2027*

- Led the design and development of TreeStruct3D, extending 3DCodeBench from renderable 3D code generation to structurally editable Blender assets with explicit part hierarchies and attachment relationships.
- Proposed Part-Attachment Trees and geometry-derived shared anchors to preserve parent-child connectivity under independent geometric edits, and developed a Blender-based validator to evaluate hierarchy, contact, anchor consistency, part coverage, and attachment robustness.
- Developed a structure-aware generation, validation, and repair framework that identifies edge-level structural failures, provides targeted feedback to vision-language models, and iteratively revalidates both code and rendered outputs before acceptance.
- Built and analyzed a benchmark from 212 Infinigen seeds, extracting 692 semantic part groups and 480 attachments; the proposed pipeline outperformed 3DCodeBench across four VLMs, achieving  $2.85\times$ – $21.00\times$  higher end-to-end editability and 97.08%–100% surface-contact survival for recovered reference edges.

### **Affordance2Action: Task-Conditioned Scene-Level Affordance Grounding for Real-Time Manipulation**

Litao Liu, Yifan Han, Pengfei Yi, Wenbo Yu, Hanqing Wang, Haoran Du, Enze Yuan, Zilin Yuan, **Ruiding Feng**, Michael Liu, Qi Zhang, Jingjin Yu

*Under review by CoRL 2026*

- Reproduced representative affordance-grounding and robot-manipulation methods to establish technical baselines and support evaluation for the A2A framework.
- Constructed A2A-Bench with 3,300+ instance-level, task-conditioned affordance masks across 900+ object-centric scenes, annotating task-specific functional regions for robotic manipulation.
- Performed systematic annotation quality control by auditing later-stage batches, refining mask boundaries and labels, and identifying instruction-to-region mismatches and ambiguous functional regions for further review.
- Contributed to A2A-AffordGen, which achieved the best performance across all reported benchmark metrics, outperforming the strongest baselines by 13.7 gIoU points on single-instance scenes and 13.6 sIoU / 19.8 P@50 points on multi-instance scenes.

## RESEARCH EXPERIENCE

### **M.S. Thesis Research — VLMs for 3D Generation**

*Intelligent Visual Interfaces Lab, Rutgers University | Advisor: Prof. Mubbasisir Kapadia* 05/2026 - 05/2027

- Investigating VLM-driven generation of Blender Python programs to create 3D assets that are visually coherent, structurally and parametrically editable, and reusable across downstream modeling and simulation workflows.
- Pursuing both harness-level agentic optimization—including iterative planning, execution, rendering, evaluation, and code repair—and post-training with SFT and LoRA to improve Blender-code correctness, structural reasoning, and cross-category generalization.

### **Patent Analysis Using Natural Language Tools**

12/2021 - 01/2022

*Research Assistant, advised by Zhaoqi Cheng, Ph.D., Boston University*

- Collected research papers and reviewed state-of-the-art models in the field.
- Collected, compiled, and tokenized patent data from the USPTO website using database tools such as SQL and natural language tools such as Python/NLTK.
- Ran topic modeling on the patent corpus using TF-IDF and LDA to extract important textual features from patent documents.
- Designed and trained a Naive Bayes model using machine learning tools such as scikit-learn, attaining 85% accuracy in predicting firm performance measures.

## PROFESSIONAL EXPERIENCE

### **iFLYTEK**

On Site

*Assistant SDE Intern*

06/2026 - 08/2026

- Integrated an LLM-wiki knowledge base into iFLYTEK's LangChain-based Harness V2 healthcare agent platform, connecting retrieval directly to agent conversations.
- Built a knowledge-base skill that enabled DeepSeek V4 to decide when supporting content should be retrieved and used to ground its responses.
- Implemented lightweight keyword-based retrieval over uploaded and edited knowledge entries without introducing a vector database dependency.
- Designed an Obsidian-inspired interface for creating, editing, and organizing structured knowledge-base content and configurable schemas.
- Managed uploads through MinIO and associated each file with its owning user and knowledge base to maintain data isolation across accounts.
- Added automated schema self-checks and locally validated upload, retrieval, editing, and response flows using simulated hospital cases.

### **Sunyard Hangzhou**

On Site

*Intern*

06/2025 - 12/2025

- Developed backend services and algorithms for an enterprise AI agent platform built on LangChain and LangGraph, supporting configurable agent workflows.
- Integrated LLaMA-Factory as a platform feature that provided configurable LoRA fine-tuning and model-adaptation functionality.
- Built data-generalization pipelines with Easy Dataset, DataFlow, and GraphGen to prepare datasets for model training and evaluation.
- Containerized DeepSeek and Qwen3 serving stacks with Docker and configured vLLM, SGLang, and TensorRT-LLM for production inference.
- Implemented backend APIs for knowledge-base analytics, token-usage monitoring, and conversation tracking across agent sessions.
- Migrated existing agent workflows from the legacy platform to the new LangChain/LangGraph-based system by mapping their configurations and dependencies.

### **NIO Shanghai**

On Site

*Intern*

02/2023 - 05/2023

- Designed Unity-based in-vehicle interactions that translated Firefly's brand identity into coherent interface behaviors and usage scenarios.
- Prototyped scenario-driven cockpit features, including a camping mode that carried the Firefly brand concept into a distinct vehicle-use experience.
- Built a test pipeline that fed real-time speed, RPM, and gear telemetry from Forza Horizon 4 into Unity to validate instrument-cluster behavior.
- Studied gasoline-vehicle driving habits to identify EV transition pain points and inform interface requirements, implementing contextual synthetic engine audio in selected driving states to provide familiar feedback.

## SKILLS

- Languages: Chinese (Native), English (Advanced)
- Technical Skills: Python, C/C++, Java; FastAPI, RAG, LLM Post-Training (SFT, LoRA/PEFT), MCP, REST APIs, Workflow Orchestration; Docker, Redis, MySQL